Nptel Online Certification Course
Indian Institute of Technology Kharagpur
Computer Vision
Assignment - Week 2

Number of questions: 10 **Total marks: 10x2=20**

QUESTION 1: **Type: MCQ**

A possible equivalent representation of a 2D point $(\frac{1}{2}, \frac{3}{4})$ in R^2 after projecting onto P^2 is

a) $(2, 3, 1)$

b) $(3, 2, 4)$

c) $(4, 6, 8)$

d) $(6, 4, 3)$

Correct Answer: c)
Detailed Solution:
It is natural to consider the set of vectors $(kx, ky, k)^T$ for varying values of k to be a representation of the point $(x, y)^T$ in R^2 after projecting onto P^2.

FOR QUESTIONS 2 AND 3:

Given a homography $H = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix}$. Based on the given data solve the following questions 2 and 3:

QUESTION 2: **Type: Comprehensive**

Find the transformation of the point $(5, 3)$.

a) $(6, -13, -10)$

b) $(13, -6, 10)$

c) $(13, 6, 10)$

d) $(13, -6, 11)$

Correct Answer: c)

Detailed Solution:

Transformation of a point x is given by $x' = Hx$.

QUESTION 3: **Type: Comprehensive**

Find the transformation of the line passing through the points $p1 = (2, 0)$ and $p2 = (1, -3)$.

a) $11.5x + 10y - 5.5 = 0$

b) $10x - 11.5y + 5.5 = 0$

c) $10x - y + 5.5 = 0$

d) $10x - 11.5y - 5.5 = 0$

Correct Answer: d)

Detailed Solution:

Line passing through the points is given by $l = p1 \times p2$.

Transformation of the line l is given by $l' = H^{-T}l$.

QUESTION 4: **Type: MSQ**

Which of the following properties are true for a homography?

a) it transforms point in real R^2 space to projective P^2 space.

b) three points lie on the same line if and only if their homographic transforms lie on the same line.

c) minimum 4 point correspondences needed to compute homography.

d) only in the form of singular 3×3 matrix.

Correct Answer: b), c)

Detailed Solution:

a) Homography is a projective transformation, i.e. $P^2 \rightarrow P^2$. It preserves collinearity, and is always in a linear form. d) A homography matrix must be non-singular (invertible).

QUESTION 5: **Type: MCQ**

Given a homography $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ Compute the transformation of dual conic C_∞^* $(I.J^T + J.I^T)$ under H, where $I = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$ and $J = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$.

a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Correct Answer: c)

Detailed Solution:

Dual Conic transformed under H can be written as $HC_\infty^*H^T$, where $C_\infty^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

QUESTION 6: **Type: MCQ**

Given a homography $H = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix}$. Find the vanishing line.

a) $(-0.5, 1, 0.25)$

b) $(-0.5, 1.25, 0.25)$.

c) $(-0.5, 0.25, 0.25)$.

d) $(0, 0, 1)$.

Correct Answer: c)
Detailed Solution:

The vanishing line is given by $\begin{bmatrix} 1 & 1 & -2 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix}^{-T} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

QUESTION 7: **Type: MSQ**

Which of the following statements are true?

a) The degree of freedom for similarity group is 4 (1 scale, 2 rotation, 1 translation).

b) The circular points are fixed points under the projective transformation H if and only if H is a similarity.

c) A conic remains a conic under homography.

d) Any square matrix can be decomposed as a product of an orthogonal matrix and an upper triangular matrix.

Correct Answer: b), c), d)

Detailed Solution:

The degree of freedom for similarity group is 4 (1 scale, 1 rotation, 2 translation).

QUESTION 8: **Type: Numeric**

Given two lines $l = 2x + y - 3 = 0$ and $m = 3x - 2y + 1 = 0$ meet at a point p. Find the Euclidean angle between these two lines. Answer should be in nearest degrees. Discard the decimal values.

Correct Answer: 60

Detailed Solution:

The Euclidean angle is given by $cos\theta = \frac{l^T C_\infty^* m}{\sqrt{(l^T C_\infty^* l)(m^T C_\infty^* m)}}$, where $C_\infty^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

QUESTION 9: **Type: MCQ**

Recollect Direct Linear Transform (DLT) algorithm for non-homogeneous equation $Ah = 0$. Choose the equation from which the matrix A is formed relating a point X_i and its transformed point $X_i^{'}$ in 2D projective spaces, where $X_i' = (x_i', y_i', w_i')^T$ and $X_i = (x_i, y_i, w_i)^T$, $i = 1, 2, \cdots n$.

a) $\begin{bmatrix} 0^T & -w_i' X_i^T & y_i' X_i^T \\ w_i' X_i^T & 0^T & -x_i' X_i^T \\ -y_i' X_i^T & x_i' X_i^T & 0^T \end{bmatrix} \begin{pmatrix} h^1 \\ h^2 \\ h^3 \end{pmatrix} = 0$

b) $\begin{bmatrix} y_i' X_i^T & 0^T & -w_i' X_i^T \\ w_i' X_i^T & 0^T & -x_i' X_i^T \\ -y_i' X_i^T & 0^T & x_i' X_i^T \end{bmatrix} \begin{pmatrix} h^1 \\ h^2 \\ h^3 \end{pmatrix} = 0$

c) $\begin{bmatrix} 0^T & -w_i' X_i^T & w_i' X_i^T \\ x_i' X_i^T & 0^T & -x_i' X_i^T \\ -y_i' X_i^T & y_i' X_i^T & 0^T \end{bmatrix} \begin{pmatrix} h^1 \\ h^2 \\ h^3 \end{pmatrix} = 0$

d) $\begin{bmatrix} 0^T & 0^T & w_i' X_i^T \\ x_i' X_i^T & 0^T & -x_i' X_i^T \\ -y_i' X_i^T & y_i' X_i^T & 0^T \end{bmatrix} \begin{pmatrix} h^1 \\ h^2 \\ h^3 \end{pmatrix} = 0$

Correct Answer: a)

Detailed Solution:

The matrix A is formed from the following equation $\begin{bmatrix} 0^T & -w_i' X_i^T & y_i' X_i^T \\ w_i' X_i^T & 0^T & -x_i' X_i^T \\ -y_i' X_i^T & x_i' X_i^T & 0^T \end{bmatrix} \begin{pmatrix} h^1 \\ h^2 \\ h^3 \end{pmatrix} = 0.$

QUESTION 10: **Type:MCQ**

Given the circle of radius 4 with centre at $(-3, 2)$ in R^2 and homography $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Which of the following represents the circle by a conic C?

a) $C = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 3 & -2 & -3 \end{bmatrix}$

b) $C = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 3 & -2 & 3 \end{bmatrix}$

c) $C = \begin{bmatrix} -1 & 0 & -3 \\ 0 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix}$

d) $C = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -2 \\ -3 & -2 & -3 \end{bmatrix}$

Correct Answer: a)

Detailed Solution:

Equation of a circle is given by $ax^2 + bxy + cy^2 + dx + ey + f = 0$. From this equation conic C is given by $C = \begin{bmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{bmatrix}$.
